Handling Data 3 — Measures of dispersion · Quartiles · Cumulative frequency

Quartiles for small data sets: Q_1 is the $\frac{1}{4}(n+1)$th value, the median the $\frac{1}{2}(n+1)$th, Q_3 the $\frac{3}{4}(n+1)$th — if that falls between two values, take their mean. Squared paper for the graph question. Marks in brackets. ★ = harder question.

1 The average number of rainy days per month in a city is: Jan 2, Feb 1, Mar 3, Apr 4, May 15, Jun 24, Jul 27, Aug 26, Sep 20, Oct 12, Nov 5, Dec 2. Calculate **a** the range, **b** the median, **c** the inter-quartile range. [6]

In order ($n = 12$): 1, 2, 2, 3, 4, 5, 12, 15, 20, 24, 26, 27.

a $27 - 1 = 26$. **b** $\frac{13}{2} = 6.5$th value: mean of 6th and 7th $= \dfrac{5+12}{2} = 8.5$ days.

c Q_1: 3.25th → between 3rd and 4th: $\dfrac{2+3}{2} = 2.5$. Q_3: 9.75th → between 9th and 10th: $\dfrac{20+24}{2} = 22$. IQR $= 22 - 2.5 = 19.5$ days — the wet and dry seasons make the middle half very spread out.

2 Mira divides her garden into nine equal squares and counts the daisies in each: 10, 5, 15, 5, 10, 0, 10, 5, 15. Calculate **a** the range, **b** the median, **c** the inter-quartile range. [6]

In order ($n = 9$): 0, 5, 5, 5, 10, 10, 10, 15, 15.

a $15 - 0 = 15$. **b** 5th value: 10. **c** Q_1 = 2.5th → $\dfrac{5+5}{2} = 5$; Q_3 = 7.5th → $\dfrac{10+15}{2} = 12.5$. IQR $= 7.5$.

3 The marks of 30 boys and 30 girls in a test are grouped below. [13]

Marks x (%)	Boys	Girls
$0 < x \leq 20$	3	1
$20 < x \leq 40$	7	3
$40 < x \leq 60$	10	8
$60 < x \leq 80$	6	11
$80 < x \leq 100$	4	7

a Draw a cumulative frequency graph for each set on the same axes, and use the graphs to estimate **(i)** the median and **(ii)** the inter-quartile range of each.

b Compare the performance of the boys and the girls.

a Cumulative frequencies (plotted at the *upper* class boundaries 20, 40, 60, 80, 100): boys 3, 10, 20, 26, 30; girls 1, 4, 12, 23, 30. Join each set with a smooth S-shaped curve from $(0, 0)$. Read across at $\frac{1}{2}n = 15$ for the median, 7.5 for Q_1 and 22.5 for Q_3. Boys: median ≈ 50, $Q_1 \approx 33$, $Q_3 \approx 68$, IQR ≈ 35. Girls: median ≈ 65, $Q_1 \approx 49$, $Q_3 \approx 79$, IQR ≈ 30. (Accept readings within a few marks.)

b The girls did better on average — their median is about 15 marks higher. Their IQR is slightly smaller, so their marks are a little more consistent; the boys' marks are more spread out.

Total: 25 marks

Check: order the data first · the IQR measures the spread of the middle 50% and ignores extremes · cumulative frequency is plotted at the *end* of each class · for large n read the quartiles at $\frac{1}{4}n$, $\frac{1}{2}n$, $\frac{3}{4}n$ · compare with an average *and* a spread.